Operating Systems

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Lecture 15: Building images to run in Docker containers. Run LAPACK in Docker

Build and run image

- docker build -t <image name> .
- docker run <image name>
- docker run -t <interactive image name>

Example of image

- FROM ubuntu:latest
- RUN /bin/bash
- CMD ls

Dockerfile

- docker build -t u1 .
- docker run u1

Listing

```
hare@hare-world:~/z$ docker build -t u1 .
Sending build context to Docker daemon  2.048kB
Step 1/3 : FROM ubuntu:latest
 ---> a2a15febcdf3
Step 2/3 : RUN /bin/bash
 ---> Running in d5c0900ebec4
Removing intermediate container d5c0900ebec4
 ---> 7ffbfbb2281b
Step 3/3 : CMD ls
 ---> Running in 1524cd305ced
Removing intermediate container 1524cd305ced
 ---> c49cd08ef60b
Successfully built c49cd08ef60b
Successfully tagged u1:latest
hare@hare-world:~/z$ docker run u1
bin  dev  home  lib64  mnt  proc  run  srv  tmp  var
boot  etc  lib  media  opt  root  sbin  sys  usr
hare@hare-world:~/z$
```

Example of interactive image

- FROM ubuntu:latest
- RUN /bin/bash

Dockerfile

- docker build -t u2 .
- docker run -it u2

Listing

```
hare@hare-world:~/z$ docker build -t u2 .
Sending build context to Docker daemon  2.048kB
Step 1/2 : FROM ubuntu:latest
 ---> a2a15febcdf3
Step 2/2 : CMD /bin/bash
 ---> Using cache
 ---> 44a70b6d4a7f
Successfully built 44a70b6d4a7f
Successfully tagged u2:latest
hare@hare-world:~/z$ docker run -it u2
root@d67dc6a4d8de:/# ls
bin  dev  home  lib64  mnt  proc  run  srv  tmp  var
boot  etc  lib  media  opt  root  sbin  sys  usr
root@d67dc6a4d8de:/# exit
exit
hare@hare-world:~/z$
```

Build and run LAPACK in Docker

I. Install C and Fortran programming systems

II. Install LAPACK package and LAPACKE interface

III. Use the example_DGESV_rowmajor.c program to test the package

IV. Compile and run the test program using the test Makefile

Layers of software and names of images

Layers	Image names
lp-ex1	myubu4
lp-ex1.c, makefile	
/bin/bash	myubu3
cblas, lapacke, lapacke examples	
/bin/bash	myubu2
lapack	
/bin/bash	myubu1
gcc, gfortran	
ubuntu	

Files and directories

- hare@hare-world:~/lpd1$ ls -R
  myubu1 myubu2 myubu3 myubu4
- ./myubu1:
  Dockerfile
- ./myubu2:
  Dockerfile lapack-3.8.0.tar.gz
- ./myubu3:
  Dockerfile
- ./myubu4:
  Dockerfile lp_ex1.c Makefile

How to use built images

Image names	How to use
myubu4	Just solve a single example SLAE and observe the results
myubu3	Interactive /bin/bash; benchmarking on various platforms using routines from lapacke examples
myubu2	Interactive /bin/bash; programming in gfortran using lapack routines
myubu1	Interactive /bin/bash; programming in gcc and gfortran

Install programming systems

1. Install compiler-linker gcc (and also g++, make)

```
sudo apt update
sudo apt install build-essential
```

2. Install compiler-linker gfortran

```
sudo apt update
sudo apt install gfortran
```

Build image myubu1

```
FROM ubuntu:latest
RUN apt-get update && apt-get install -y \
    build-essential \
    gfortran \
    python
RUN /bin/bash
```

Dockerfile

- docker image build -t myubu1 .
- docker run -it myubu1

Listing of myubu1

```
hare@hare-world:~/lpd1/myubu1$ docker build -t myubu1 .
Sending build context to Docker daemon  2.048kB
Step 1/3 : FROM ubuntu:latest
 ---> a2a15febcdf3
Step 2/3 : RUN apt-get update && apt-get install -y      build-essential
gfortran      python
 ---> Using cache
 ---> e1d8f743018f
Step 3/3 : RUN /bin/bash
 ---> Running in 64e26f367545
Removing intermediate container 64e26f367545
 ---> 95f478500b62
Successfully built 95f478500b62
Successfully tagged myubu1:latest
hare@hare-world:~/lpd1/myubu1$ docker run -it myubu1
root@0202079b45c9:/#
```

Install package LAPACK

1. Download the lapack-3.8.0.tar.gz archive with the LAPACK source code from

   http://www.netlib.org/lapack

2. Unzip the archive to your home directory

   tar -zxvf lapack-3.8.0.tar.gz

3. Change to the LAPACK directory

   cd lapack-3.8.0

4. Copy the example package build options file

   cp make.inc.example make.inc

5. Edit the package build options file

   gedit make.inc

6. Build and test the LAPACK libraries

   make

Build image myubu2

```
FROM myubu1:latest
COPY lapack-3.8.0.tar.gz .
RUN tar -zxvf lapack-3.8.0.tar.gz
RUN cp /lapack-3.8.0/make.inc.example /lapack-3.8.0/make.inc
RUN ulimit -s unlimited && cd /lapack-3.8.0 && make
RUN /bin/bash
```

Dockerfile

- docker image build -t myubu2 .
- docker run -it myubu2

Listing of myubu2

```
…
-->   LAPACK TESTING SUMMARY  <--
Processing LAPACK Testing output found in the TESTING directory
SUMMARY                 nb test run numerical error    other error
================        =========== ================== ================
REAL                1292645           0      (0.000%)     0      (0.000%)
DOUBLE PRECISION    1293457           0      (0.000%)     0      (0.000%)
COMPLEX             752816      0     (0.000%)     0      (0.000%)
COMPLEX16                752536      1      (0.000%)     1      (0.000%)

--> ALL PRECISIONS      4091454           1      (0.000%)     1
      (0.000%)

Removing intermediate container 2d9910e61eb3
 ---> 39da51af1dca
Step 6/6 : RUN /bin/bash
 ---> Running in 3603da68889c
Removing intermediate container 3603da68889c
 ---> 80192fe29ba9
Successfully built 80192fe29ba9
Successfully tagged myubu2:latest
```

Install interfaces LAPACKE and examples

7. Build the CBLAS interface
8. Build the LAPACKE interface
9. Run the LAPACKE test cases

Build image myubu3

```
FROM myubu2:latest
RUN cd /lapack-3.8.0/CBLAS && make && cd /lapack-3.8.0/LAPACKE && make && cd /lapack-3.8.0/LAPACKE/example && make
RUN /bin/bash
```

Dockerfile

- docker image build -t myubu3 .
- docker run -it myubu3

Listing of myubu3

```
hare@hare-world:~/lpd1/myubu3$ docker build -t myubu3 .
Sending build context to Docker daemon  2.048kB
Step 1/3 : FROM myubu2:latest
 ---> 80192fe29ba9
…
LAPACKE_dgels (col-major, high-level) Example Program Results
 Solution
   2.00   1.00
   1.00   1.00
   1.00   2.00
Removing intermediate container d9db78567001
 ---> 6b353bf03acd
Step 3/3 : RUN /bin/bash
 ---> Running in abea03c9fa14
Removing intermediate container abea03c9fa14
 ---> f099c004c109
Successfully built f099c004c109
Successfully tagged myubu3:latest
hare@hare-world:~/lpd1/myubu3$
```

Using myubu3 to run tests

```
root@48d12612cdb0:/# time /lapack-3.8.0/LAPACKE/example/xexample_DGESV_colmajor -n 3
 Entry Matrix A
   0.34  -0.11   0.28
   0.30   0.41  -0.30
  -0.16   0.27  -0.22
 Right Rand Side b
   0.05
  -0.02
   0.13
LAPACKE_dgesv (row-major, high-level) Example Program Results
 Solution
  -0.48
   1.19
   1.22
real    0m0.002s
user    0m0.002s
sys     0m0.000s
root@48d12612cdb0:/#
```

Compile and run test program

1. Exit the example directory, create your own test directory, and navigate to it.
2. Create the pl_ex1.c program file and Makefile using the gedit text editor, copying their contents from Appendices 1 and 2.
3. Run the test and verify the resulting solution.

Build image myubu4

```
FROM myubu3:latest
RUN mkdir /lapack-3.8.0/LAPACKE/myex
COPY lp_ex1.c /lapack-3.8.0/LAPACKE/myex
COPY Makefile /lapack-3.8.0/LAPACKE/myex
RUN cd /lapack-3.8.0/LAPACKE/myex && make
CMD /lapack-3.8.0/LAPACKE/myex/lp_ex1
```

Dockerfile

- docker image build -t myubu4 .
- docker run myubu4

Listing of myubu4

```
hare@hare-world:~/lpd1/myubu4$ docker build -t myubu4 .Sending build context to Docker daemon
5.632kB
Step 1/6 : FROM myubu3:latest
 ---> f099c004c109
Step 2/6 : RUN mkdir /lapack-3.8.0/LAPACKE/myex
 ---> Running in 6b80bd0f0056
Removing intermediate container 6b80bd0f0056
 ---> 908c015b4950
Step 3/6 : COPY lp_ex1.c /lapack-3.8.0/LAPACKE/myex
 ---> e92748d6c522
Step 4/6 : COPY Makefile /lapack-3.8.0/LAPACKE/myex
 ---> 592bc9d81cc8
Step 5/6 : RUN cd /lapack-3.8.0/LAPACKE/myex && make
 ---> Running in f7d7b9325f14
gcc -O3 -I. -I../include -c -o lp_ex1.o lp_ex1.c
gfortran  -o lp_ex1 lp_ex1.o ../../liblapacke.a ../../liblapack.a ../../librefblas.a
./lp_ex1
-1.313167e+00
-1.000000e-01
1.800000e+00
Removing intermediate container f7d7b9325f14
 ---> 3db494237364
Step 6/6 : CMD /lapack-3.8.0/LAPACKE/myex/lp_ex1
 ---> Running in 942936e3184e
Removing intermediate container 942936e3184e
 ---> d27389b678e4
Successfully built d27389b678e4
Successfully tagged myubu4:latest
```

Run myubu4

```
hare@hare-world:~/lpd1/myubu4$ docker
run myubu4
0.301587
-0.873016
2.095238
hare@hare-world:~/lpd1/myubu4$
```

Use LAPACK to benchmark Docker and other container systems (hypervisors) on various platforms